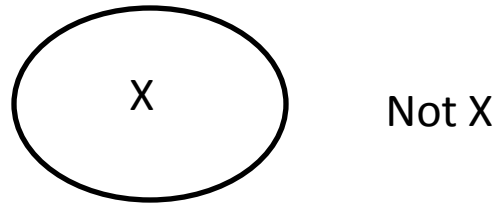


COMMUNICATION AND DECISION MAKING

ISSS Saturday Morning
September 5, 2020

Robert Johansson

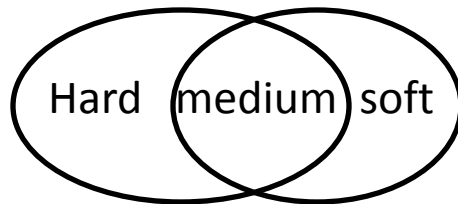
Information is choice within a set, or a decision.



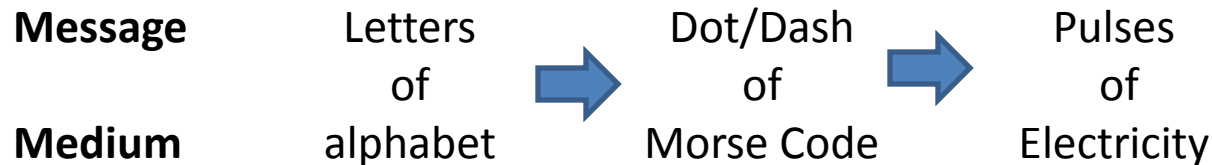
This is the basis of categorical logic.

For the neurological perspective see McCullough & Pitts on
How the Brain Perceives Universals.

The categorical nature of information allows for dialectical logic:
thesis, antithesis, synthesis,
and an increasingly differentiated perception.



Communication is the translation of choices in one medium into choices in another medium.



Since information is a decision, it can also function as a command, or program.

For example

translating the alphabet into Morse code or the alphabet into binary



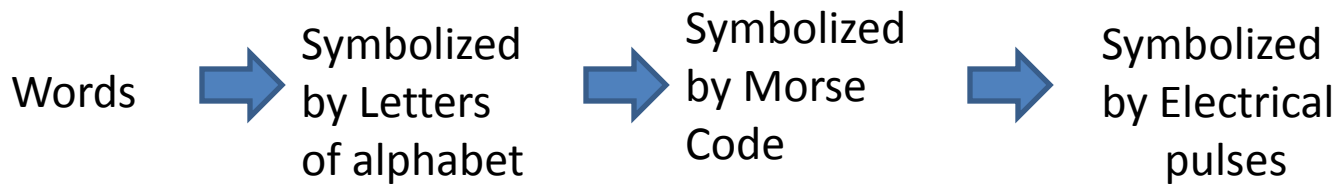
A = dot dash
B = dash dot dot dot
C = dash dot dash dot
Etc.



A = 000000
B = 000001
C = 000010
Etc.

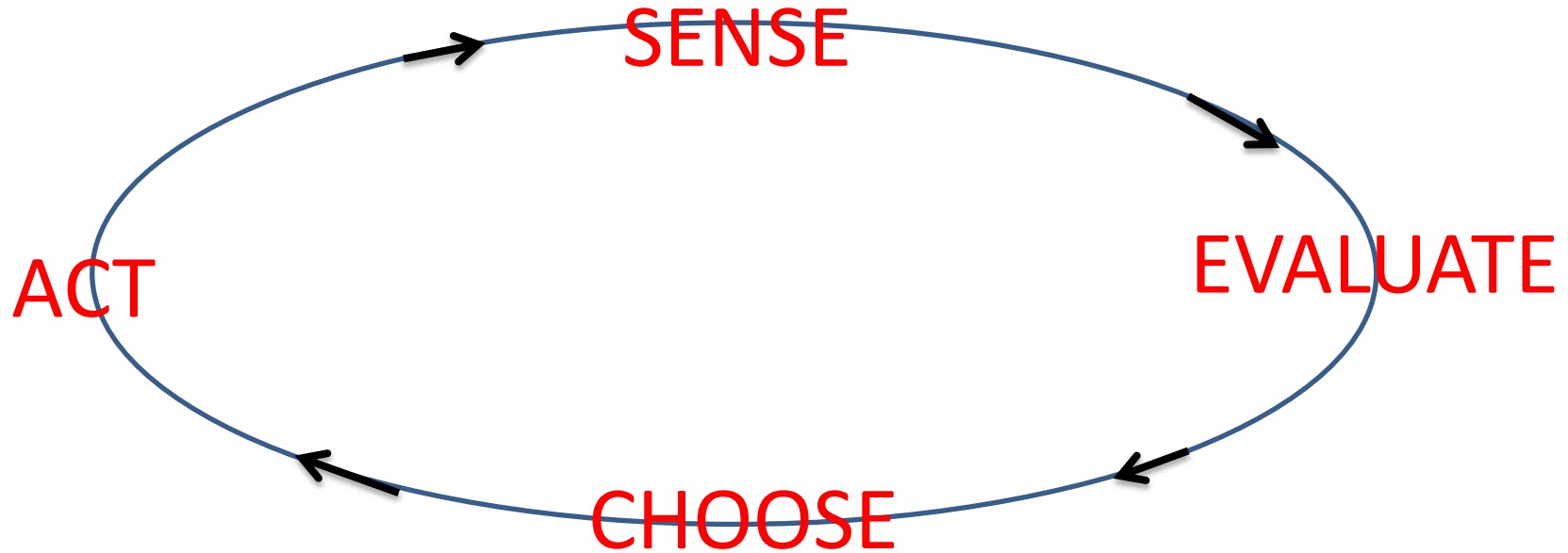
These logical commands can also function as a sorting process.

Communication because it is a process of translation creates symbolic relationships. Communication can be a series of symbolic translations.



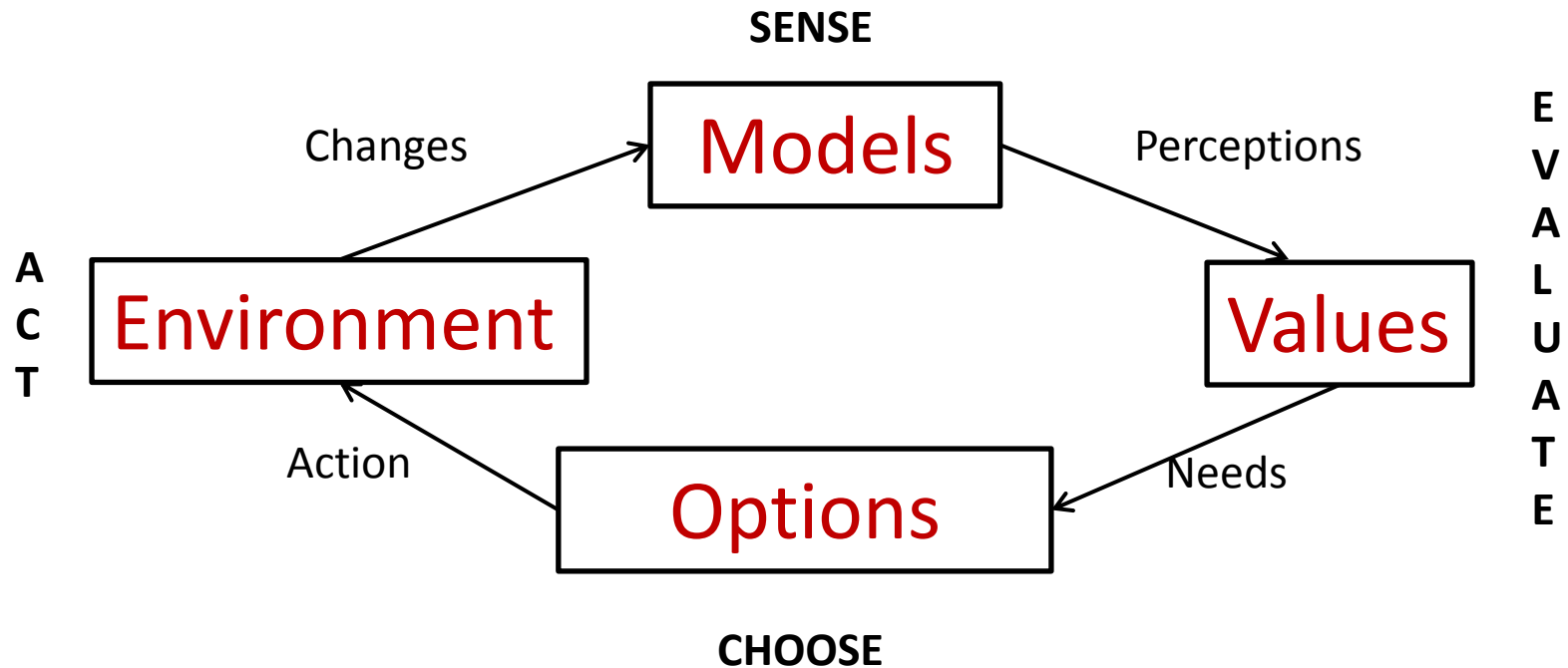
Contrary to naïve empiricism our perception of an object is not the object. The map is not the territory. The perception is an abstract symbol of the object. The object is the meaning of the perception. As the territory is the meaning of the map. Meaning is a relationship.

The Cybernetic Control Loop



These four processes form a loop. Any control process consists of going around this basic control loop.

A Cybernetic Control System is a Communication Process



Models, values, options and environment are the translation processes that constitute the program that controls the control process.

Conscious decision making takes the process to another meta-level and consists of programming the translators: listing options, defining criteria, researching the possible results.

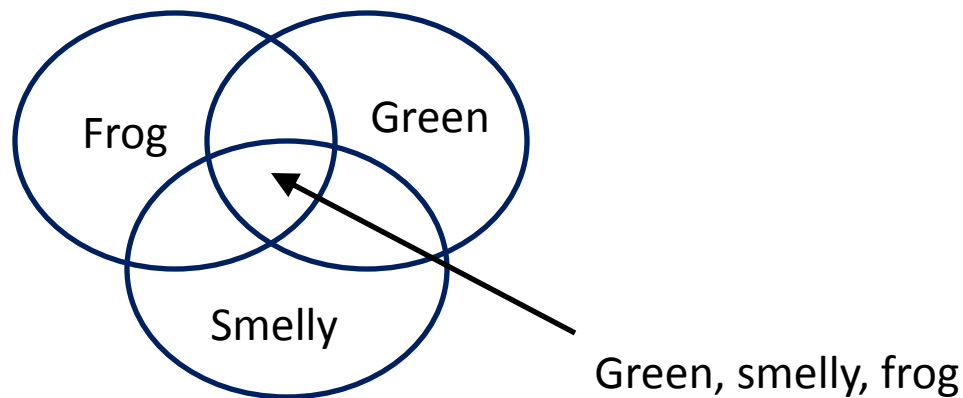
The environment communicates with decision making systems through their sensors.

The object in the environment forms the medium into a message.

The decision system's sensor translates the message into a perception.

<u>Medium</u>	Translator	<u>Message</u>	Translator	<u>Perception</u>
Light	Frog	Colour wave	Eye	green frog
Chemicals	Frog	Scent	Nose	Smelly frog

The application of categorical differentiation allows the object to be perceived as a whole with properties. The message is perceived as a property of the object.



The feedback process is essentially harmonic. It is attempting to create a harmony between the environment and a set of values.

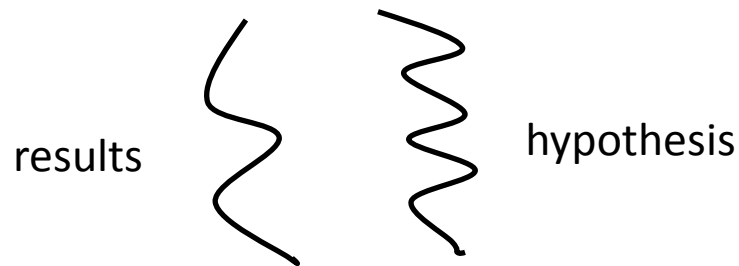


The lack of harmony is experienced as cognitive dissonance, which requires resolution. In the way that G7 demands a C major.

Feedback gives us meta information.

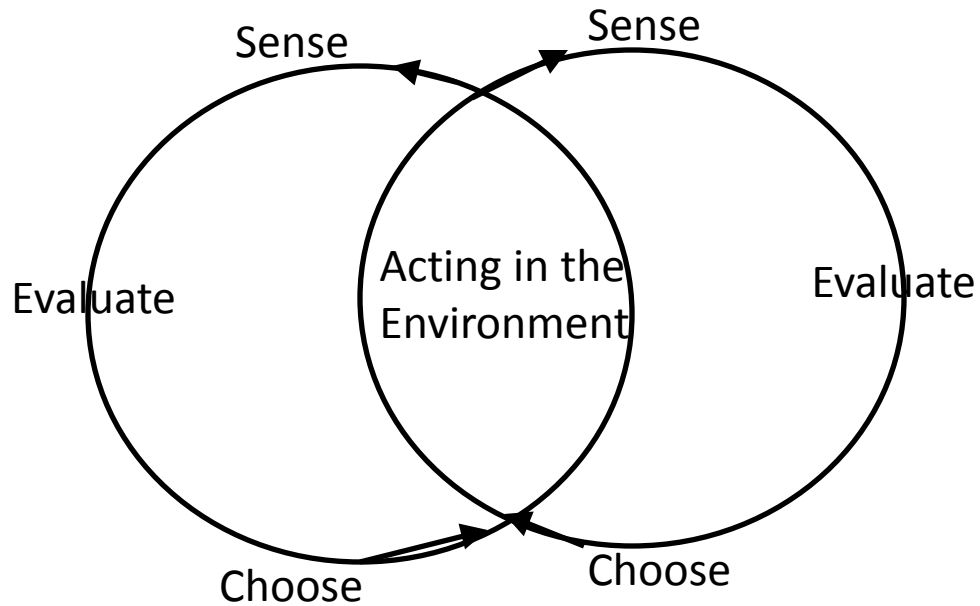
Because we control the action, the results of the action not only give information about the effectiveness of the action, but also about the accuracy of the model.

This is the scientific method – the results of the experiment are used to evaluate the accuracy of the hypothesis.



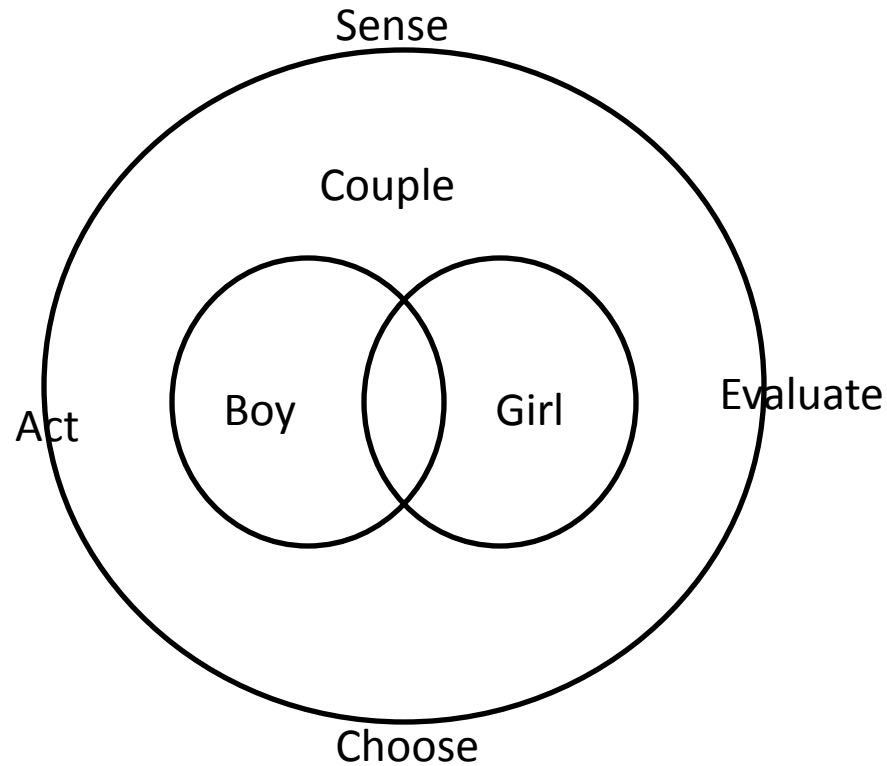
When Decision Making Systems Meet

They communicate.



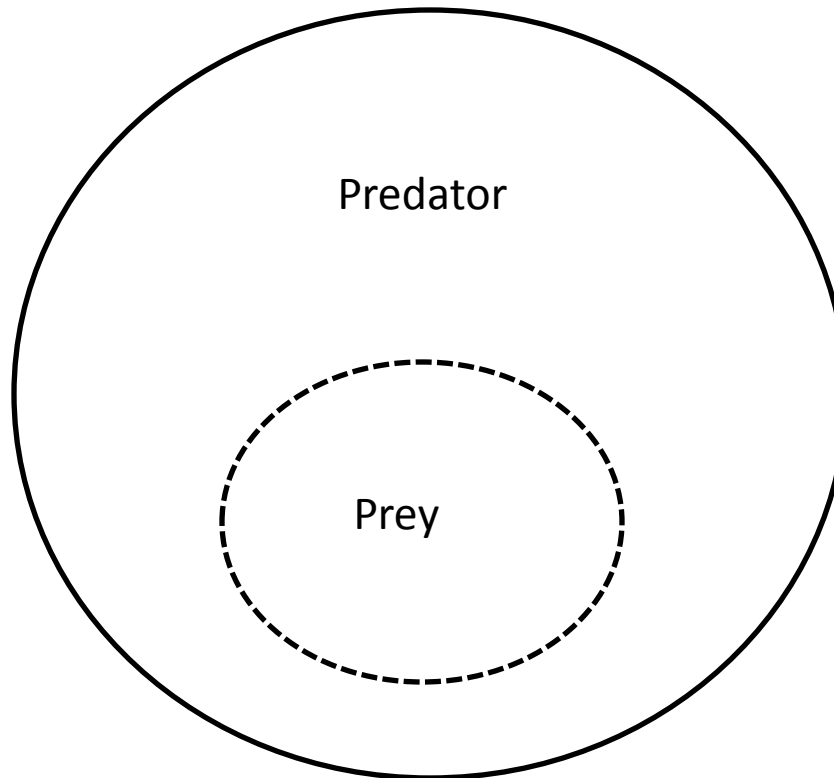
They become part of a shared environment.

They form a larger control system that contains them as parts.



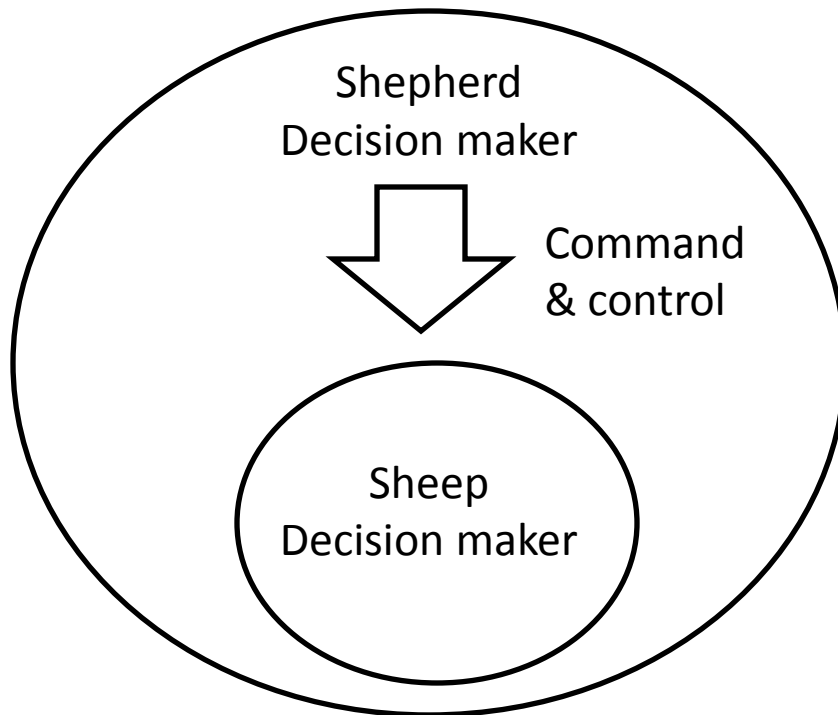
When Smuts coined the term “holism” this was the evolutionary process he was describing, and what Koestler called “holarchy.” How does the larger whole make decisions for the community?

Absorption



In the predator/prey relationship the prey organism is broken down and the useful bits absorbed into the predator. It no longer has a decision function.

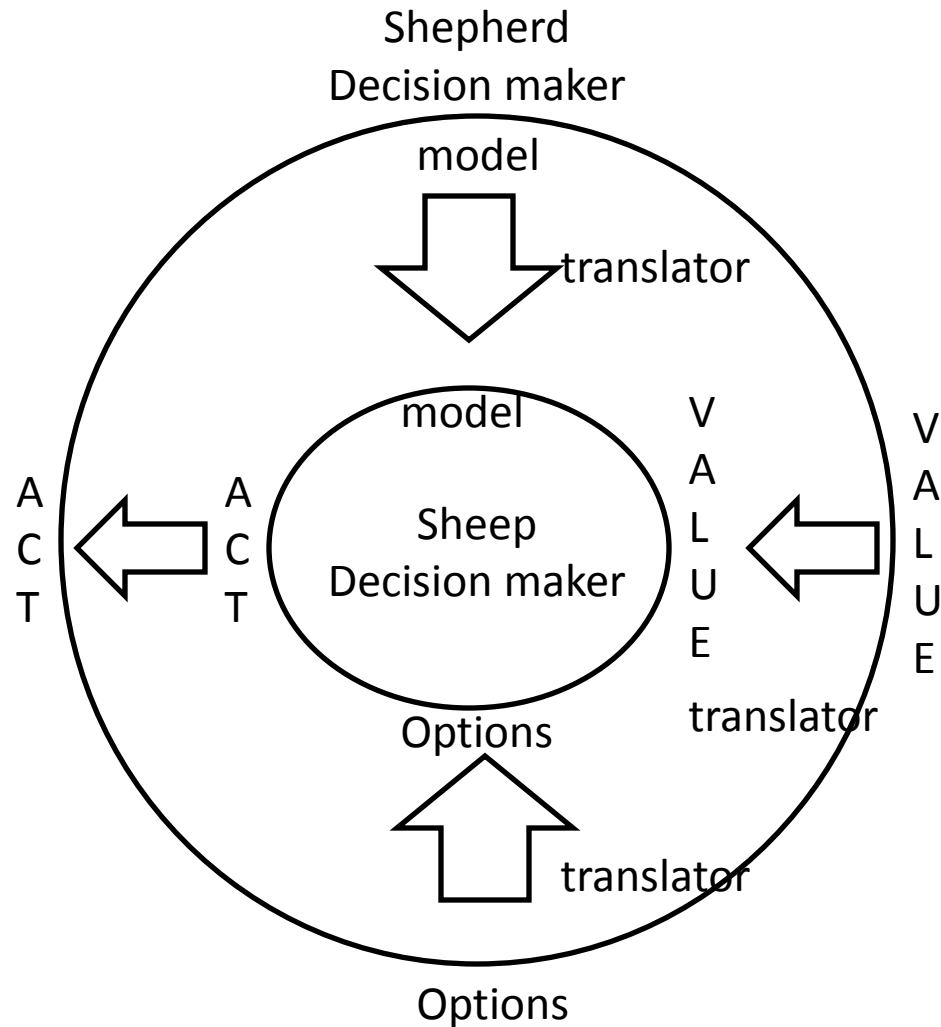
Dominance



Master – slave
Husband – wife
Management – Labour
Teacher – student
Captain – crew

In dominance systems the dominant one makes the decisions for the subordinate. The communication becomes one of command and control. Since the subordinate remains a decision making system control is exercised by control of the programming (models, values and options).

Command
& control



The shepherd has to translate his model into sheep language in order to communicate with the sheep: Restricting options by using a chute, rewarding appropriate behaviour, using a Judas goat to lead it into the truck. The action of the sheep becomes part of a coordinated action of shepherd and flock.

A Common Language

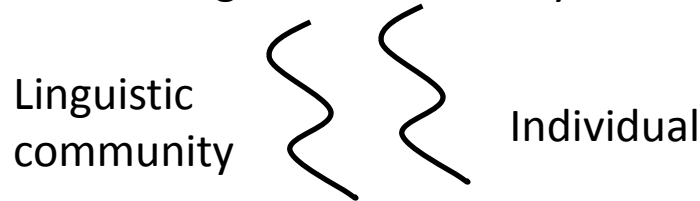
The parts of a whole and the levels of a holarchy have to communicate, and to do that they need a common language. The ideas expressed by the whole have to have a meaning in the context of the part.

The trainer gives the dog a command in English. When the dog obeys the command the trainer gives the dog a treat. The command is in English, and the dog translates whatever that sounds like to a dog into a command.

Building a Common Language

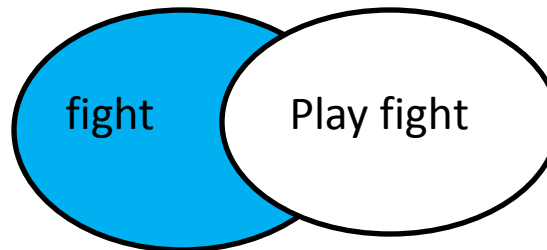
Communication is about harmonizing -- the matching of the message to the meaning. We each fit a message into our understanding. But we test the fit by action and response, the feedback process.

Linguistic communication is harmonic. The goal is for the receiver to understand the same meaning as the transmitter. Building a language is a harmonic goal that strives for the same symbols to have the same meaning for each member of the linguistic community that uses the language.

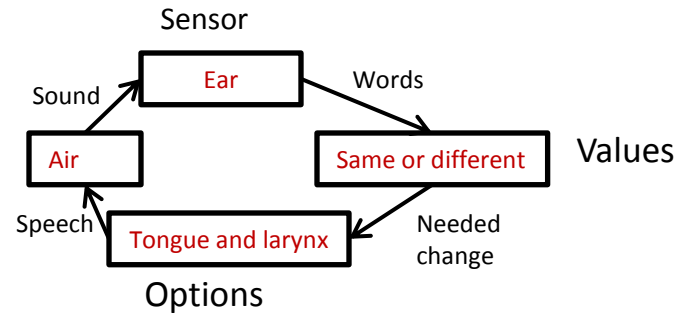


Species have a variety of communication modes. The bees use the dance. There are specific symbols for mating, dominance, territoriality, learning.

The meta-language. Because information is categorical it allows for a meta-level. Thus Bateson discovered that young Otters could tell the difference between a fight and play fight.



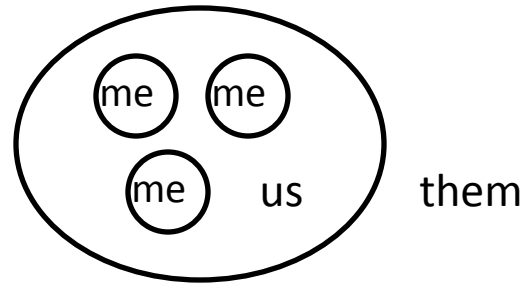
Building a common language is a feedback process that is constantly ongoing.



A child's first word is usually "dada" or "mama," a word that makes the fundamental phonemic distinction between a consonant and a vowel. Contrary to popular opinion, a phoneme is not a sound, but a set of distinctions between sounds.

The growing fluency of the child is a growing ability to distinguish between more subtle differences between sounds and to reproduce mimetically the same distinctions of sound. And eventually the individual learns the distinctions between the 20,000 to 35,000 words in the average vocabulary.

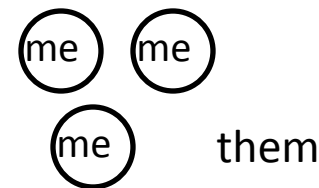
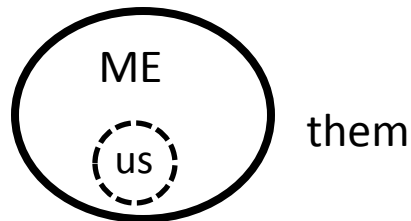
The creation of a group identity.



Integration: Any group has a boundary that is the difference between us and them, the insiders and the outsiders, those who belong and those who don't. It is the identity of the group as a whole.

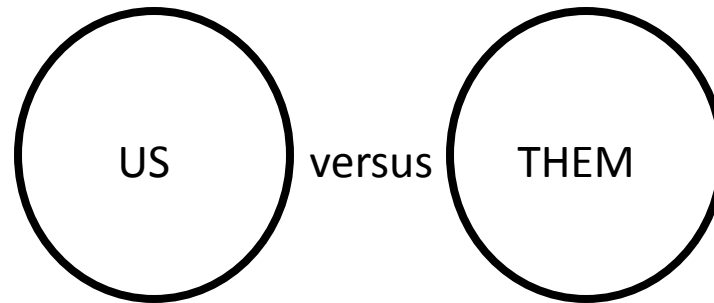
With a couple this is fairly simple. With larger groups this can be a fuzzy boundary. The outsider becomes an observer, becomes a participant, becomes a worker, becomes a decision maker, becomes an authority. This process can be defined in terms of physical distancing, and can be conceived of as drawing closer to the centre. With a couple these gradations of belonging can be psychological. In terms of the distance metaphor couples grow closer together or further apart.

Disintegration:



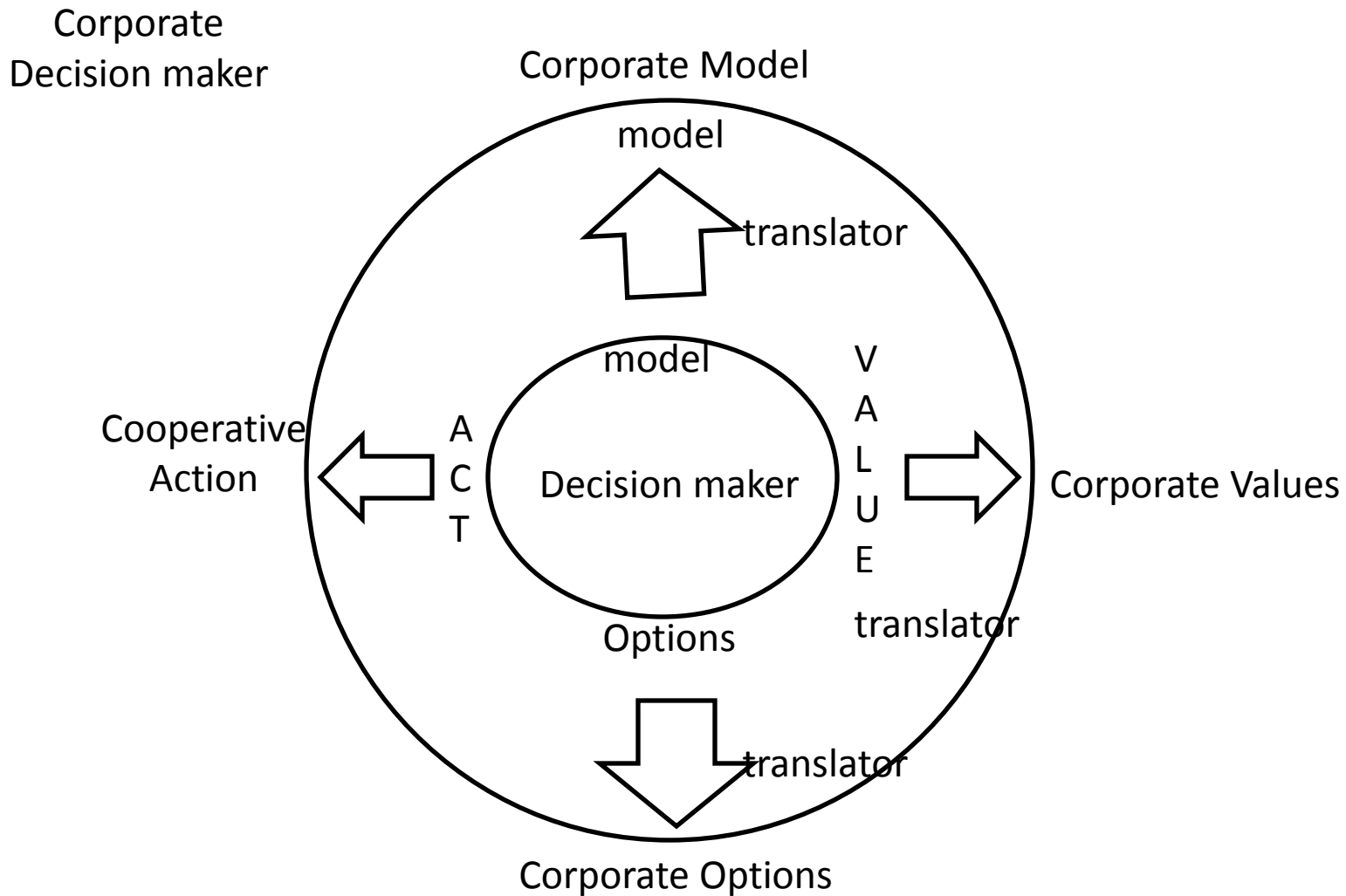
Anarchy and Libertarianism seek to make the common identity a function of the ego and thus weaken the common identity and cause it to disintegrate.

Schismogenesis

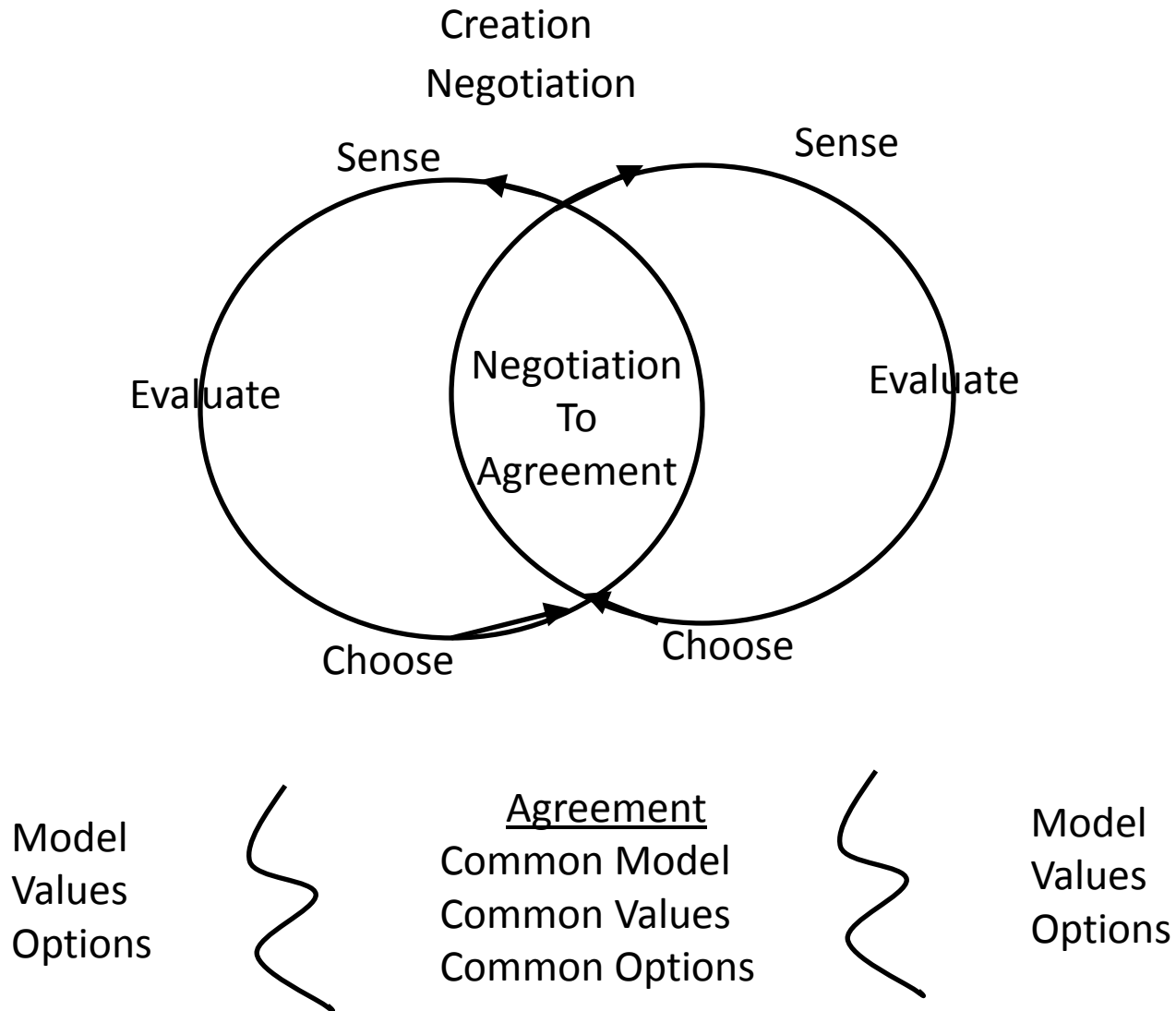


The term “schismogenesis” was coined by Bateson to describe the process of increasing differentiation between groups. Thus for some people the definition of a Canadian is “not an American.” Sometimes this goes to self-destructive extremes.

Because the difference is a part of the identity the boundary becomes very rigid. One of the commonest expressions of this is in racism.



The new whole will develop a common model, common values, common options, and in order to achieve cooperative action.

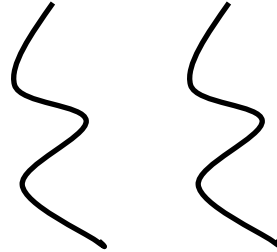


The model, values and options of the two parties come to an agreement.

Initiation

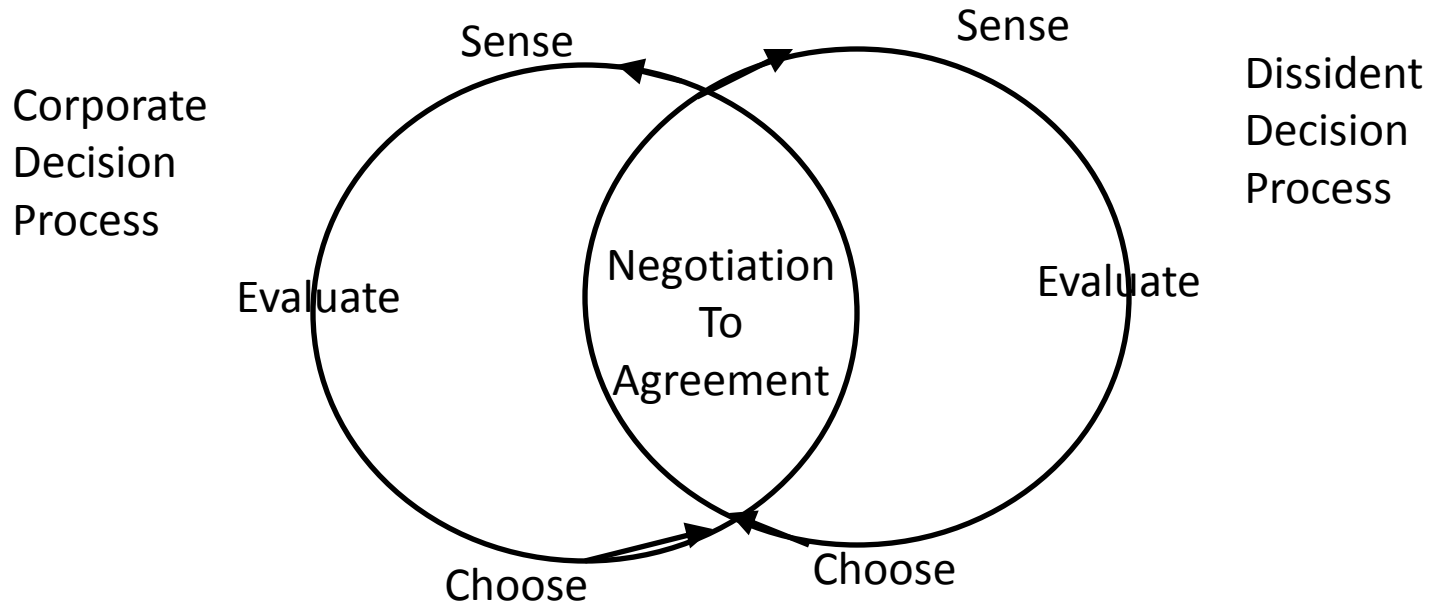
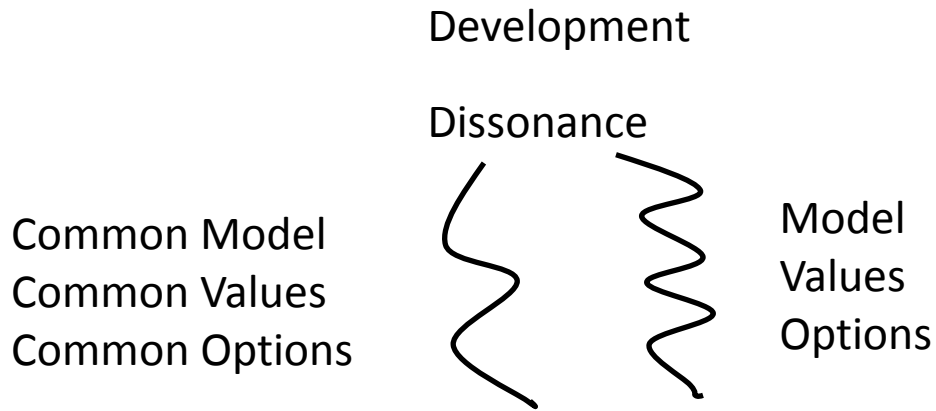
Community

Common Model
Common Values
Common Options



Individual

Model
Values
Options



The Road to Agreement

We are taught in the popular media that the road to agreement is through force. It generally involves beating people up or killing them. There is another way.

Appreciative Process - common models

The first step is usually the appreciative sharing of models. Often the disagreements people have come from different models of the world. When people listen to each other then they can see things from a different point of view. It creates a common model of the world that contains both points of view.

The Influence Process – building common values

The sharing of different values generates a common values space. It expands the range of considerations that the group has to recognize if a common value is to be reached.

The Control Process – building common actions

Once there is a shared model and a common set of values, then the alternatives available can be assessed in a way that recognizes the needs of each member of the community.

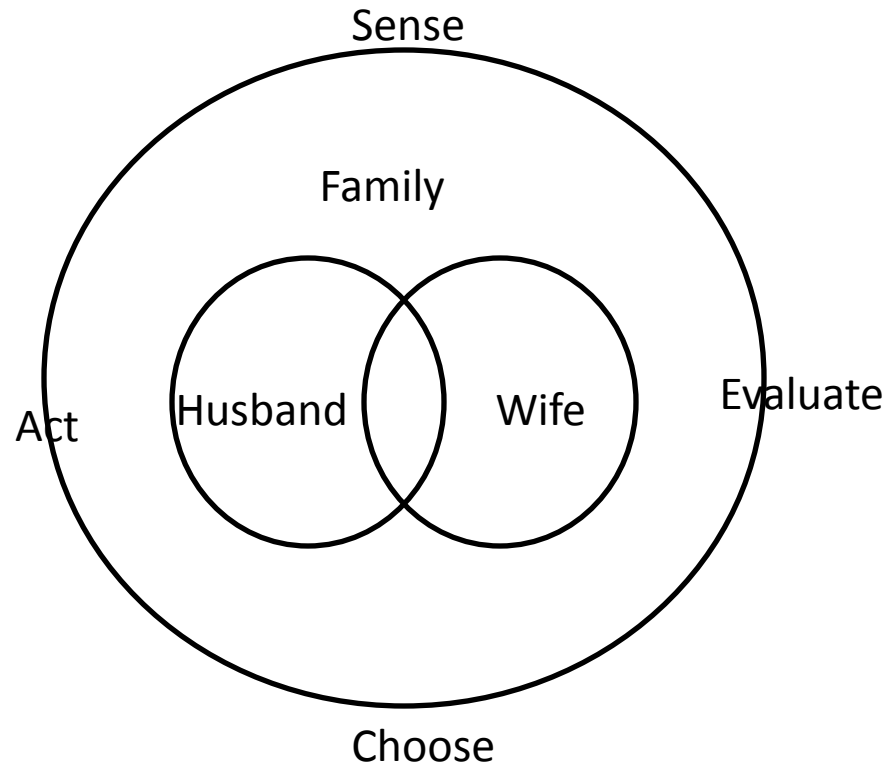
Leadership

The agreement gives the group authority and the individuals responsibilities. England expects this day every man to do his duty. But the individuals are decision makers and must respect the authority.

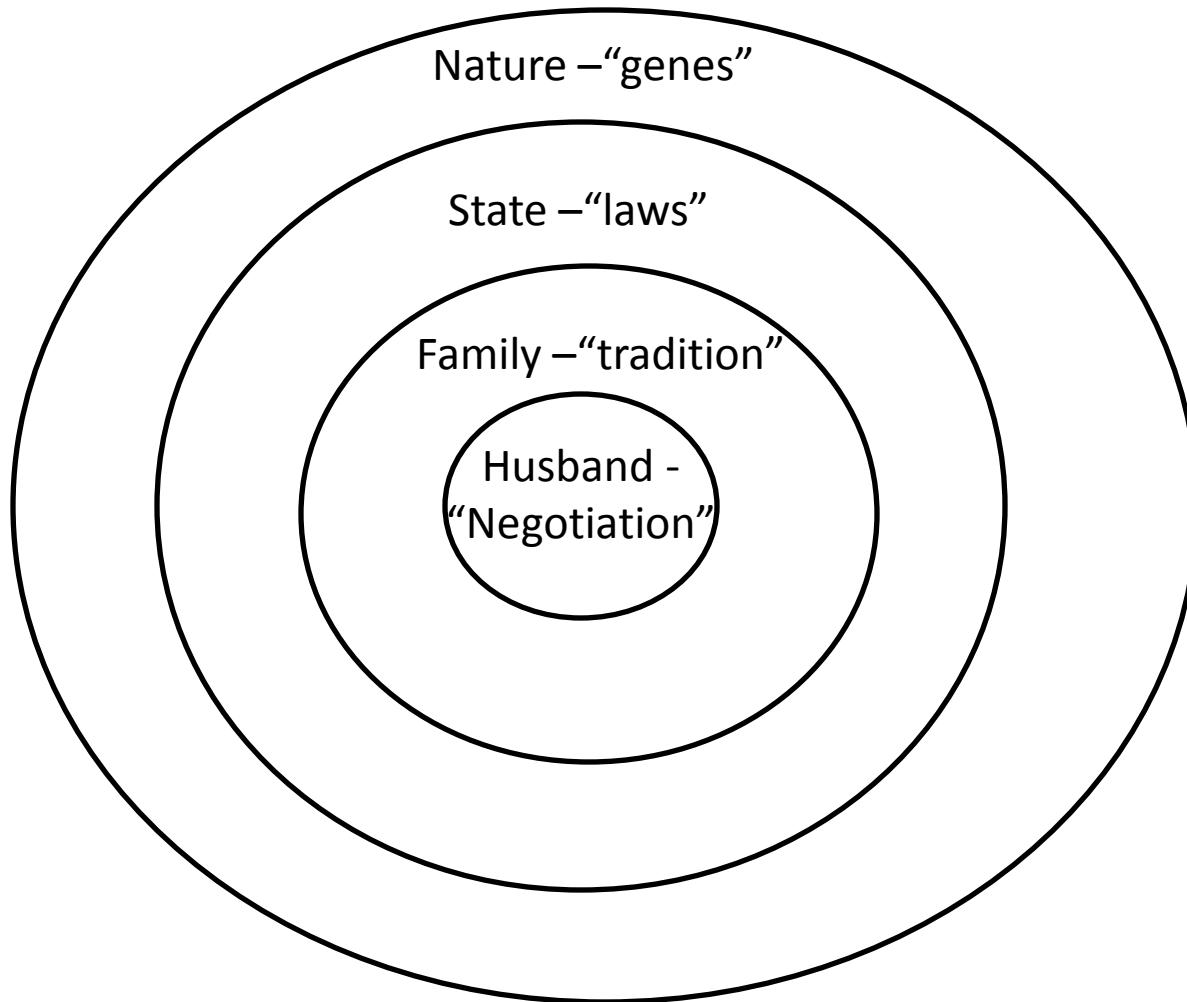
The army attempts to match leadership and authority. The officer has the right to give commands, but will the men follow this officer.

Roles

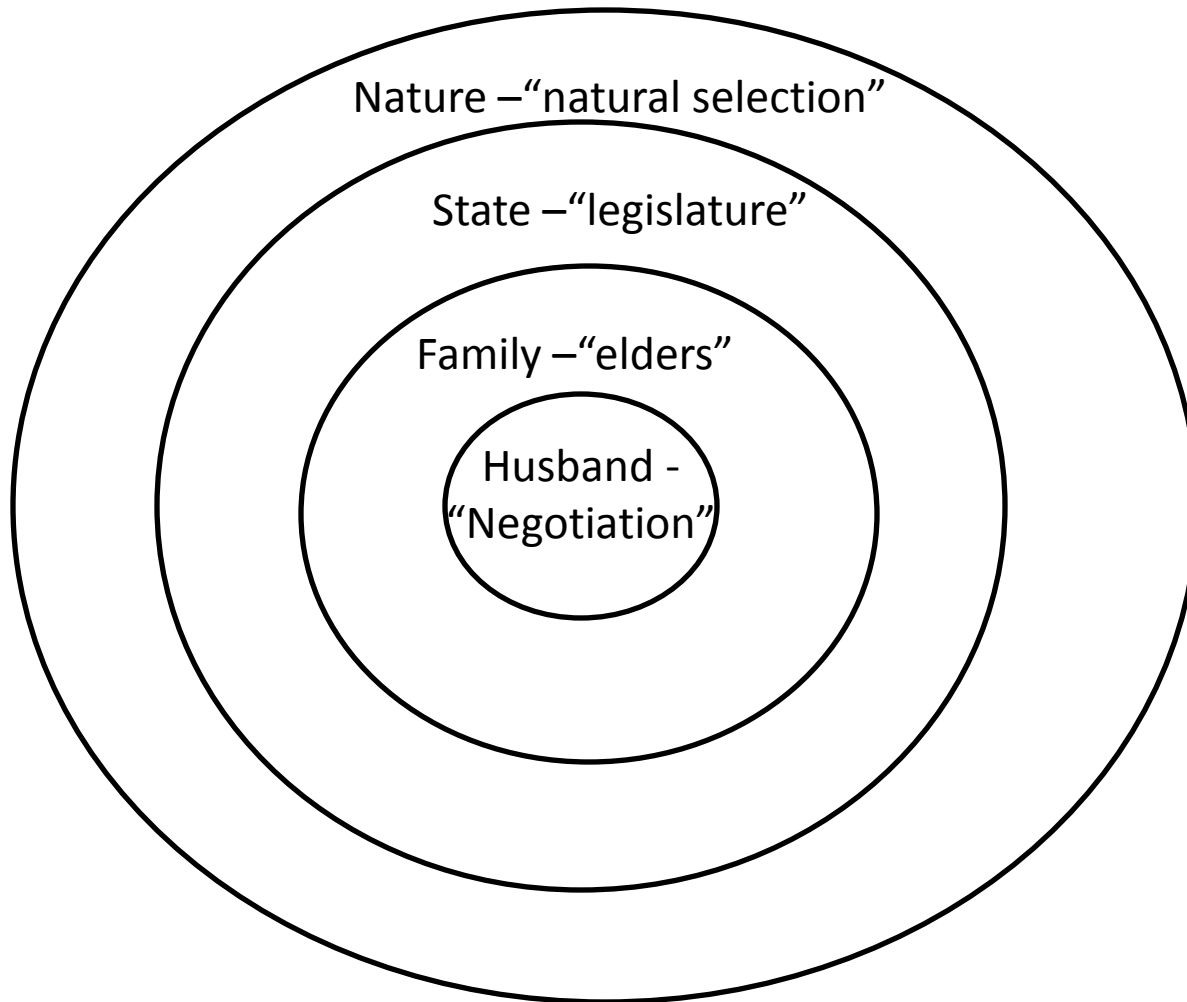
In complex modern organizations authority and responsibility are often confined to specific roles. The functioning of the organization becomes the ability of different teams to communicate and cooperate.



These roles are frequently defined by larger wholes, the holarchy.
Each with its own medium of communication.



Each with its own decision making process.



Systemic Racism

When is racism a function of individuals and when is racism a function of “the system.” Is it authorized behaviour?

Is race part of the corporate model of the world?

It is important that there be a recognition of race. This is good racism. Race is a major factor in the modern world. It needs to be seen.

What is the place of race in the corporate value set?

Are racial differences valued? Is there racial preference? Is it appropriate? Attempts to achieve a racial balance can be appropriate. The problem is that the culture is historically racist.

Is the corporate technology racist?

Application forms. Job descriptions. Evaluation. Supervision. Promotion. Customer service. Racial profiling and red-lining are notorious examples.

Are the results racist?

There is no question that the results of the justice system are racist. It is an indicator that there is a racial bias in the processes. Even if this is a cultural bias it can still be located and dealt with.